

Cheng-Han Lin

Full-stack developer · Taichung, Taiwan

Computer science graduate and full-stack developer with hands-on experience building production LLM systems — retrieval-augmented generation, self-hosted model inference and machine-learning forecasting — alongside the web, mobile and payment infrastructure around them. I ship complete products solo and care about evaluation, privacy and the failure cases most people skip.

Projects

HeartBox — AI mood-journaling platform (RAG + emotion forecasting)

Solo developer · Capstone project · heartbox.tw · github.com/alanlin0604/HeartBox

Designed, built and deployed a mental-health platform end to end, combining retrieval-grounded LLM feedback with a machine-learning emotion forecast. Sole developer across ML, backend, frontend, mobile and infrastructure. **Live demo at heartbox.tw — sign in with test1 / test1, no registration required.**

- **RAG pipeline** — BGE-M3 embeddings over a ChromaDB vector store built from seven clinical sources (WHO, APA, NHS, NIMH). Retrieval triggers when an entry's sentiment falls below -0.4 and returns the top-3 relevant passages, which the model must cite — so advice is traceable to a source rather than hallucinated.
- **Random Forest forecasting** — engineered 53 features (12 behavioural and physiological metrics $\times$ four lag windows, plus five calendar features) over 22,796 training rows. 5-fold cross-validation: MAE 0.22 on sentiment (range -1 to $+1$), MAE 1.04 on the stress index (range 0–10), and AUC 0.948 with 88% recall on high-stress-day classification over 31,720 rows — deliberately tuned for recall, since a missed warning costs more than a false alarm.
- **Model selection** — evaluated linear regression, a single decision tree, Random Forest, XGBoost and an LSTM against data requirements, interpretability, inference cost and overfitting risk. Chose Random Forest for sub-50 ms CPU inference that trains on only a few hundred rows per user while staying interpretable through feature importance.
- **Self-hosted inference** — ran open-weight models behind a FastAPI GPU service exposed through a Cloudflare Tunnel: LLaVA-v1.6-Mistral-7B for vision, and TAIDE-LX-7B, an open-weight model tuned for Traditional Chinese. Journal text stays inside a controlled environment instead of going to a third-party API.

- **Privacy and safety** — journal contents encrypted with Fernet (AES-128-CBC + HMAC-SHA256) using environment-injected keys, plus 2FA. Three-step separated consent modelled on GDPR Art. 7 and Taiwan’s PDPA, with AI-training consent independently refusable and parental confirmation required for users aged 13–17. Crisis-keyword detection across journals, AI chat and the anonymous community surfaces national helplines immediately.
- Also delivered: PHQ-9 / GAD-7 assessments, mood trend and correlation visualisations, sleep and habit analytics, a 103-achievement system, and an Android build via Capacitor.

LapseWatch — subscription renewal reminder service

Solo developer · Jun 2026 – present · [lapsewatch.smallworks.app](#)

- Built a service that notifies users **before** subscriptions auto-renew, through LINE, email and desktop notifications, with Google Calendar sync and CSV / JSON export.
- Developed a Chrome extension that detects subscriptions automatically, reading only a narrow window of text around matched keywords to minimise data collection.
- Integrated a complete subscription billing flow with statutory e-invoicing (ECPay, Taiwan’s e-invoice system); designed privacy-first, with no bank account linking and no cross-site tracking.

PantryKeeper — shared household inventory tracker

Solo developer · Jun 2026 – present · [pantrykeeper.net](#)

- Built a multi-user inventory and expiry tracker with per-member permission controls, offline support and waste statistics.
- Wrote a bulk import parser that accepts pasted data from Notion, Excel and Google Sheets, inferring quantity and unit from free-form text.

Skills

Area	Detail
Languages	Python, JavaScript, TypeScript, SQL, HTML, CSS
AI / ML	Retrieval-augmented generation, LangChain, ChromaDB, embedding models (BGE-M3), self-hosted LLM inference, scikit-learn, Random Forest, feature engineering, cross-validation, model evaluation and selection
Backend	Django, Django REST Framework, Django Channels (WebSocket), FastAPI, Node.js, REST API design, PostgreSQL

Area	Detail
Frontend	React, Vite, Tailwind CSS, Recharts, Chrome extension development
Infrastructure	Google Cloud Run, Cloudflare Pages and Tunnel, Capacitor (Android), Git, CI/CD
Security	Fernet / AES encryption, JWT, 2FA, rate limiting, GDPR and Taiwan PDPA consent design
Integrations	Payment gateway and statutory e-invoicing (ECPay, Taiwan), LINE Messaging API, Google Calendar API

Education

National Chin-Yi University of Technology — Taichung, Taiwan M.S. in Computer Science and Information Engineering · from September 2026

National Chin-Yi University of Technology — Taichung, Taiwan B.S. in Computer Science and Information Engineering · graduated June 2026

Capstone project: HeartBox (above) · Student Association, Administration Division

Languages

Mandarin Chinese (native) · English (professional reading and writing)

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